



10 AI Trends Revolutionising Manufacturing Industry

Artificial intelligence (AI) has been playing a crucial role in the growth of manufacturing industries by boosting productivity and efficiency. The article below explores some of the trends that AI technology is bringing in the modern manufacturing industries like smart factories, 3D printing and blockchain. These trends ensure more flexibility and improve speed and quality of products



Representation of artificial intelligence (Credit: www.freightwaves.com)

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A smart manufacturing unit is different from a traditional manufacturing unit in many ways. A typical modern manufacturing unit is not a clumsy industrial space with a few or thousands of human workers or robotised production units under one roof. With the advancement in technologies, such as artificial intelligence (AI), machine learning and the Internet of Things (IoT), there is much more than the traditional manufacturing

processes and setups. AI is being employed in manufacturing to make businesses more efficient by extending human capabilities. There are many areas where AI is helping to improve the processes of manufacturing industries and boost productivity and efficiency. This article covers some of the AI trends that have revolutionised the modern manufacturing industry including smart factories, 3D printing, AI-robots, generative designs and blockchain.

1. Smart factory

Automation in manufacturing or factory is not a new thing. A smart factory is based on the modern industrial automation concept. It is a highly digitised and connected production facility, considered as an outcome of the fourth industrial revolution or Industry 4.0. The smart factory employs modern technologies such as AI, robotics, analytics, Big Data and the IoT. With the use of modern technologies, the smart factory systems can learn and adapt in real-time, enable more flexibility in operation and run without much human intervention. The characteristics of the smart factory include visibility, connectivity, and autonomy. Asset management, supply chain management and inventory management are some of the hottest areas of AI, machine learning and IoT adoption in a smart factory.

2. AI and 3D printing

3D printing is mainly used in rapid prototyping, design iteration, and small-scale